

Transparency and Legitimacy: The Norwegian General Pension Fund Global and Long-Term Sovereign Wealth Performance

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Abstract:

Over the past few decades, sovereign wealth funds have become formidable global financial actors, collectively managing trillions in assets globally. Yet a fundamental question persists: what institutional mechanisms enable some funds to deliver sustained macroeconomic benefits whilst others fail to insulate wealth from political pressure or short-term fiscal demands? This paper investigates how transparency in sovereign wealth fund governance affects core economic objectives: risk-adjusted financial performance, intergenerational wealth preservation, and macroeconomic stabilisation, through the case of Norway's Government Pension Fund Global (GPFG). We develop and test a transparency-legitimacy-outcomes framework, arguing that transparency operates by generating political and economic legitimacy that protects long-term viability, sustains adherence to fiscal rules, and reduces governance complications. Using a single-case, time-series design we determine that transparency does not deliver short-term outperformance, but delivers credibility and discipline necessary for funds such as the GPFG to preserve equity risk capacity through market downturns. This research contributes to the literature on sovereign wealth fund governance by empirically linking transparency practices to measurable economic outcomes through the mediating rule of legitimacy, in order to understand the economic consequences of institutional design and political legitimacy in state-owned investments.

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1 Introduction

Since the 1950s, sovereign wealth funds have emerged as major actors in the global financial system. Initial funds, such as the Kuwait Investment Authority in 1953 and the Abu Dhabi Investment Authority in 1976, were created as a mechanism for resource-rich nations to transform finite natural resource wealth into diversified financial assets that could sustain national prosperity beyond resource depletion. Yet, this resource dependency was deemed fragile by the 1970s and 1980s due to heightened exposure to commodity price volatility (Clark et al., 2013). Besides serving as national savings vehicles, sovereign wealth funds came to be understood also as instruments of intergenerational equity, macroeconomic stabilization, and long-term national development (Clark et al., 2013). By the 1990s and 2000s, the rationale for sovereign wealth funds expanded to encompass a broader range of policy objectives, such as foreign exchange reserve management, strategic economic development, and development assistance domestically and internationally. Countries such as Singapore pioneered the use of sovereign wealth funds as instruments for strategic economic development, channeling capital toward sectors deemed critical for national competitiveness and regional influence (Amar & Lecourt, 2023). Similarly, several Middle Eastern sovereign wealth funds began incorporating development objectives into their mandates, investing in infrastructure, technology transfer, and capacity building both within their home countries and across the developing world, notably in Africa and Asia (Zhang, 2016). As such, sovereign wealth funds came to occupy an increasingly ambiguous position between purely commercial investment vehicles and instruments of state policy, raising questions about their motivations, governance, and ultimately, their legitimacy in both domestic and international contexts (Amar & Lecourt, 2023; Clark et al., 2013)

The scope of these ambitions to reconcile sovereign wealth fund governance and legitimacy is largely captured in the distinctive model of the Norwegian Government Pension Fund Global (GPFG). Established in 1990 as the Government Petroleum Fund, functioning largely as the aforementioned investment vehicle to ensure intergenerational equity for Norway's petroleum wealth, the fund's creation was also rooted in a conscious political decision to avoid the "resource curse" that had afflicted other oil-rich nations (Clark & Monk, 2010). Such a curse set an empirical precedent in which sudden resource wealth led to economic volatility, inflation, corruption, and short-sighted spending, leading to the economic and political downfall of countries such as Venezuela and Nigeria (Caner & Grennes, 2009). The Norwegian Parliament instead established a fiscal rule to ensure that petroleum revenues would be deposited into the fund and invested exclusively in international financial markets, with only the expected real return -- initially set at four percent annually -- available for government spending (Caner & Grennes, 2009).

Having grown to over USD \$1.4 trillion by 2024, the Norwegian Government Pension Fund Global (GPF) has since become the world's largest sovereign wealth fund and represents a remarkable achievement in long-term wealth accumulation (NBIM, n.d.). The fund also draws particular attention for its exceptional degree of transparency and public accountability, having consistently achieved a perfect score of 100 by the Global Pension Transparency Benchmark ("Global Pension Transparency Benchmark", n.d.). The fund operates under a clear ethical mandate that includes both negative screening criteria to exclude investments in negatively received companies, with detailed annual reporting on investment decisions, risk management, and other notable decisions of the portfolio (Clark & Monk, 2010). Norges Bank Investment Management, the operational arm responsible for day-to-day investment management, publishes comprehensive quarterly reports detailing portfolio composition, geographic and sectoral allocation, performance attribution, and risk exposures (NBIM, n.d.). Furthermore, the fund is subject to regular parliamentary oversight, independent ethical council reviews, and extensive public debate within Norway about everything from specific divestment decisions to the appropriate balance between financial returns and ethical considerations (Clark & Monk, 2010). Under fiscal rule, the fund also operates with the government so that structural (cyclically adjusted) non-oil fiscal deficit, representing net government spending financed by transfers from the GPF, should over time equal approximately 3% of the fund's total value, symbolising fiscal discipline (Norwegian Ministry of Finance, 2017)

The Norwegian model raises the question: does the exceptional transparency that characterises the GPF contribute to its ability to achieve its core economic objectives, and if so, through what mechanisms does this relationship operate? Namely, it asks whether and how governance practices -- specifically the disclosure of investment strategies, portfolio holdings, and other defining frameworks -- affect a sovereign wealth fund's capacity to fulfill its economic mandate. For the Norwegian fund, these economic objectives included generating competitive long-term financial returns that can support public expenditure across generations; preserving and growing the value of accumulated resource wealth, petroleum in this case, to ensure intergenerational equity; providing macroeconomic stabilization by smoothing government spending and preventing resource-driven economic volatility; and maintaining operational capacity to pursue long-term investment strategies that may be otherwise unavailable to funds facing short-term political interference or constrained market access (Caner & Grennes, 2009; Clark & Monk, 2010; NBIM, n.d.). These objectives are defined namely to encompass broader economic outcomes that determine whether resource wealth ultimately benefits or harms national prosperity.

This paper thus addresses these questions through a comprehensive examination of how transparency in sovereign wealth fund governance affects the achievement of core economic objectives, with particular attention to the role of political and economic legitimacy as a

mediating mechanism. The primary research question guiding this study is: how does transparency in sovereign wealth fund governance contribute to the fund's ability to achieve its core economic objectives -- such as financial performance, intergenerational wealth transfer, and macroeconomic stabilisation -- and to what extent is this relationship mediated by the political and economic legitimacy that transparency generates? Through this study, we will specifically address the following:

- I. whether and how transparency contributes to the financial performance of sovereign wealth funds, including risk-adjusted returns, portfolio resilience, and long-term wealth accumulation; and
- II. whether transparency enables funds to better achieve non-return objectives, such as intergenerational equity and macroeconomic stabilization by providing policy stability.

The contribution of this paper to the existing literature on sovereign wealth fund governance is threefold. Firstly, while previous scholarship have addressed sovereign wealth fund transparency and performance as separate topics, there have been limited analysis connecting governance practices to economic outcomes. Scholars have documented the GPFG's exceptional transparency and its impressive economic track record, but rarely examine whether and how these two characteristics are causally related (Caner & Grennes, 2009; Clark & Monk, 2010). Similarly, while the literature has theorized about legitimacy as a concept, it doesn't address legitimacy's concrete economic consequences, or how political acceptance can translate into active economic results (Clark et al., 2013). This paper will treat legitimacy not as an end in itself, but as an intermediate variable with measurable economic consequences. Secondly, by comparing the Norwegian case with alternative governance models, such as with less transparent funds that still perform strongly (Zhang, 2016) and challenges of transparent funds in achieving their objectives (Caner & Grennes, 2009), this study will assess a wider range of transparency's contribution to economic effectiveness. This analysis reveals that the transparency-legitimacy pipeline is contingent on institutional context, where transparency is related to other social and political elements. Thirdly, this paper hopes to make a broader contribution to understanding the economic consequences of institutional design and political legitimacy in state-owned investments. For policymakers designing or reforming sovereign wealth fund governance frameworks, this analysis offers practical guidance: transparency should be calibrated not according to abstract best practice standards, but according to real, measured effectiveness.

2 Literature Review

2.1 Legitimacy and Its Implications

Legitimacy is defined as the social and political acceptance of an institution's authority to act (Clark et al., 2013). For sovereign wealth funds, legitimacy has rather concrete economic

consequences: funds that lack legitimacy often face political resistance, regulatory obstacles, restricted access, and other effects that ultimately diminish their ability to fulfill economic mandates (Clark & Monk, 2010). Amar and Lecourt (2023) posits that transparency is required “to maintain legitimacy domestically and credibility abroad”, establishing also a notable tradeoff: domestic legitimacy may reward discretion in pursuing industrial policy objectives, whilst external legitimacy from host countries and international markets typically prizes transparency and depoliticization of investment decisions. This framework seems to suggest that the transparency-legitimacy-economic performance pipeline is neither linear nor universally applicable, as priorities of legitimacy vary across funds and relate differently to specific economic objectives.

Clark and Monk (2010) focus mainly on the theoretical perspectives behind fund transparency, namely through their concept of “epistemic proceduralism”, arguing that transparent procedures, and publicly defensible decision-making processes constitute a direct source of legitimacy that better supports long-term economic effectiveness. In their analysis of the GPFG, they posit that transparent procedures generate social acceptance independent of investment outcomes, therefore providing political insulation that enables longer horizons for investment. Analysis by the International Monetary Fund reinforces this perspective, noting that “transparency can improve and discipline management and reduce the risk of corruption and mismanagement”, whilst simultaneously building trust in policies based on saving public assets (IMF, 2013). The theoretical implication is that transparency affects economic outcomes through multiple channels: one, by improving governance quality and reducing costs; and secondly, by building the political consensus necessary for policy stability and operational autonomy. However, while transparency may improve governance quality and reduce costs, it may also constrain operational flexibility, reveal strategies to competitors, or impose compliance burdens (Megginson & Fotak, 2015; Bernstein et al., 2013). It is thus here that legitimacy emerges instead as a mediating variable that channels transparency’s effects toward economic outcomes (Baker, 2009; Scherer et al., 2013).

Building on these insights, we follow a framework in which transparency contributes to economic objectives primarily through legitimacy rather than through direct technical channels. The logic follows: comprehensive transparency in fund governance, including financial disclosure, decision-making processes, and other responsible investment activities, generates political and social legitimacy by demonstrating accountability, enabling informed oversight and aligning fund operations with societal values (Clark & Monk, 2017). This legitimacy, in turn, produces operational insulation that protects long-term investment strategy from short-term political pressures, supports adherence to fiscal rules that enable macroeconomic stabilisation, and sustains public acceptance of the portfolio to maximise intergenerational wealth transfer (Capapé & Guerrero, 2013). From this, we can therefore compress this logic into two primary hypotheses:

H1: Transparency-driven governance contributes to consistent, modest positive returns over time, by enabling strategic discipline and reducing politically motivated conflicts of interest from optimal investment policy.

The literature on sovereign wealth performance suggests that while active management by large investors faces challenges, funds with strong governance may achieve modest but persistent value-added through several channels (Dahlquist & Ødegaard, 2018; Ang et al., 2014). This means that we don't expect there to be a strong correlation between national governance and financial returns of the fund; namely, we argue that this is attributed to research demonstrating that there are far more risks in the market to contend that there is a consistent correlation between the two. Malkiel (2003) also supports this: transparency and governance cannot overcome fundamental market efficiency in liquid markets; large positive alpha would contradict efficient market theory. We predict that transparent governance allows the fund to gain small, persistent advantages, such as crisis resilience, a more stable long-term viability, and possibly less room for error.

H2: Transparency sustains political legitimacy for maintaining investment strategies and limiting fiscal withdrawals, which results in cumulative investment returns exceeding withdrawals over time and thereby achieving intergenerational wealth.

This premise rests upon one important argument of the Solow economic growth model and other economic literature fueling policy objectives in the modern day. Exhaustible resource wealth belongs equally to current and future generations, thus creating an ethical imperative to preserve resource rents rather than consuming them entirely in the present-day (Hartwick, 1977; Solow; 1986). Applying this, we thus require that as sovereign wealth funds grow over time, natural resource stocks, such as Norway's initial petroleum reserves, will be transformed into permanent financial capital.

The political economy literature identifies some collective actions problems that undermine intergenerational equity objectives (Venables, 2016). Future generations cannot participate in current political processes, which create systematic bias toward present-oriented policies. This thus creates the famous "resource curse", in which, documented across many resource-rich countries, resource windfalls create powerful political pressures for immediate spending on visible projects that benefit current voters, dissipating resource wealth through consumption rather than investment (Humphreys et al., 2007). However, we argue that transparency regarding investment strategy: disclosure of equity allocations, return volatility, and expectations, builds public understanding of why equity risk is worth taking to achieve intergenerational objectives. Norway's comprehensive disclosure of volatility and return history has enabled political consensus around the equity-oriented strategy despite periodic market drawdowns.

3 Methodology

3.1 Assessing Previous Empirical Methodologies and Measurement Challenges

A primary empirical strategy in assessing sovereign wealth fund transparency and performance is using cross-sectional correlational analysis using transparency indices. Cuervo-Cazurra et al. (2023) employ the Linaburg-Maduell Transparency Index, which “identifies ten principles of SWF principles to the public, giving one point of transparency of each”, to measure disclosure practices across funds. Recent systematic reviews find that well-governed SWFs demonstrate better performance on its economic outcomes, namely in financial stability and long-term performance, whilst “governance weaknesses contribute to financial volatility and inefficiencies”. Aizenman and Glick (2009) correlates individual fund transparency scores with home country governance indicators from the World Bank’s Worldwide Governance Indicators, revealing that countries with high democracy ratings tend to have funds with high transparency scores.

While informative, correlational approaches face a few challenges. The positive association between transparency and performance may reflect reverse causality (successful funds have greater capacity to invest in disclosure infrastructure) or omitted variable bias (both transparency and performance may be jointly determined by underlying factors, such as resource wealth in the case of the GPFG). Amar and Lecourt (2023) address this by estimating a fractional response model to investigate whether governance disclosure norms are motivated by external or internal legitimacy, finding that “SWFs have better governance when they originate from democratic countries with high-quality national governance”.

Comparative case analysis offers deeper insight into causal mechanisms, but sacrifices generalizability. Clark and Monk (2010) employ detailed institutional analysis to demonstrate how comprehensive reporting enables parliamentary oversight, builds domestic political consensus, and insulates the fund from political pressure -- all mechanisms through which transparency plausibly contributes to intergenerational wealth preservation and fiscal stability.

3.2 General Approach

To assess the primary research question, this study employs an in-depth single-case research design focusing on the Norwegian Government Pension Fund Global. We will combine quantitative time-series analysis, longitudinal financial decomposition, and qualitative content analysis to trace the specific causal pathways linking transparency, legitimacy, and economic outcomes in what is widely recognized as the world’s most transparent sovereign wealth fund. Rather than employing cross-sectional comparison across multiple funds with varying transparency levels, this design leverages temporal variation within a single context to isolate mechanisms and therefore observe how transparency-driven legitimacy operates over time to

enable consistent achievement of financial performance, macroeconomic stabilisation, and intergenerational wealth transfer objectives.

The purpose for this design is that it enables detailed process tracing of causal mechanisms that cross-sectional analysis cannot appropriately capture, particularly the specific pathways through which transparency and legitimacy may translate into, and interact with operational autonomy, fiscal discipline, and long-term orientation (George & Bennett, 2005). Also, focusing on the GPFG -- which represents the best recognized case of transparency-driven sovereign wealth fund governance -- provides a “most likely” case for observing the theorised relationships (Gerring, 2007). Following this, the availability of comprehensive longitudinal data for the GPFG, including continuous time-series of returns, withdrawals, risk metrics, and governance disclosures over two decades, enables temporal analysis that cross-sectional designs cannot achieve.

We organise this around three main analytical components, each addressing a specific facet of the transparency-legitimacy-performance relationship and a specific economic objective. Firstly, we will employ time-series graphical analysis to assess financial performance, examining whether transparency-enabled political insulation allows superior and consistent excess returns. Secondly, we’ll use statistical correlation analysis to test legitimacy as mediator and examine its relationship to portfolio stabilisation. Because legitimacy is hard to quantify, we follow the governance and political economy literature by operationalising legitimacy using observable institutional behaviors that plausibly reflect sustained public consent and credibility. Specifically, we’ll proxy legitimacy with adherence to Norway’s fiscal rule (measures as absolute deviations from the 3% non-oil deficit benchmark), and the World Bank’s Worldwide Governance Indicators (WGI) (IMF, 2013; Clark & Monk, 2010; Aizenman & Glick, 2009). To conduct the statistical correlation testing between our various variables, we will calculate the Pearson correlation coefficient, which measures the strength and direction of the relationship between two variables, represented in the following equation as x and y , with n representing the sample size:

$$r = \frac{[n(\sum xy) - (\sum x)(\sum y)]}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

Equation 1. Pearson correlation coefficient equation

Thirdly, we’ll apply time-series analysis of the fund’s market value growth rates to quantify intergenerational wealth transfer, demonstrating how accountability enables wealth accumulation.

3.2 Data

Fund Performance and Portfolio Data are obtained from Norges Bank Investment Management’s comprehensive annual reports, published consistently since 1990 and available publicly through the NBIM website. These reports provide detailed disclosure of annual real returns for the fund

and its benchmark index; monthly return data, which will allow for return volatility calculations; portfolio composition by asset class; and other notable variables. This highly-comprehensive data suits longitudinal analysis very well.

4 Results

The following visualisation provides empirical confirmation of our previous non-linear hypothesis. We overlay the Norwegian Governance Quality Index, measured by the Worldwide Governance Indicators, onto a time-series following the GPFG's annual real returns, which represent our dependent variable of financial performance, since inception.

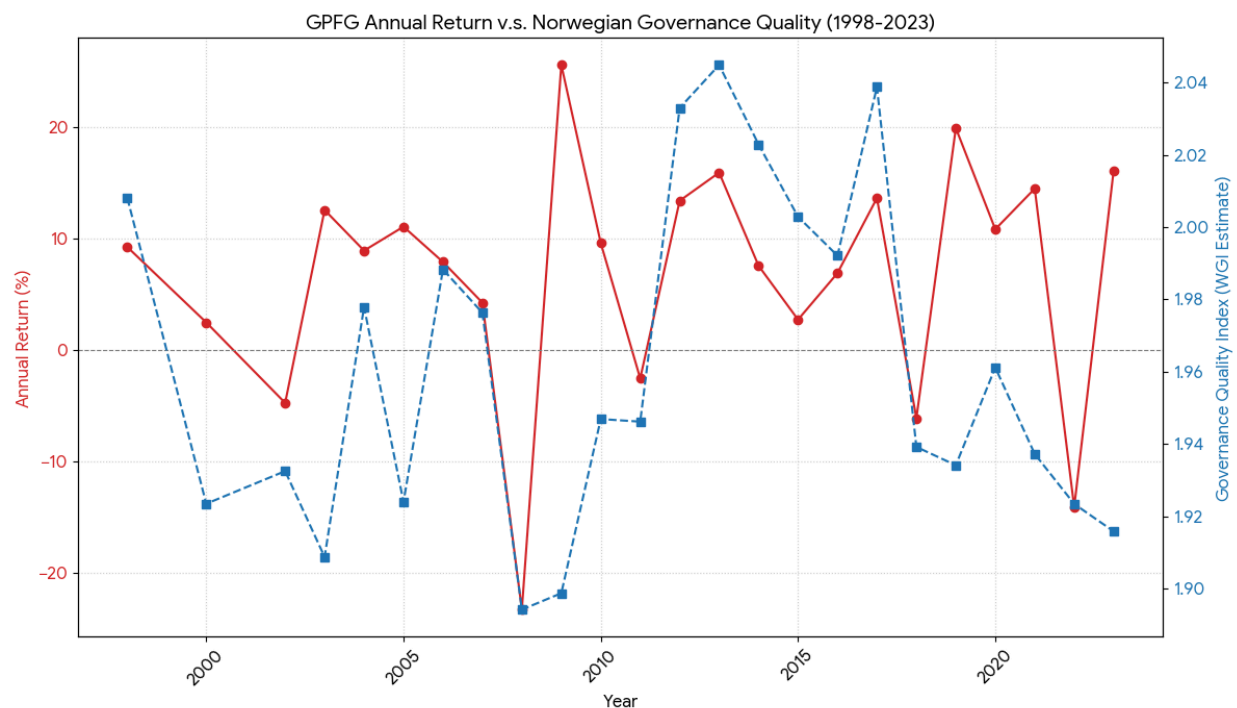


Figure 1. Time Series of Annual Real Financial Returns (%) of the GPFG v.s. Governance Quality, measured by the WGI, since inception. Data from NBIM.no and the World Bank's Worldwide Governance Indicators.

As shown, the GPFG's annual returns exhibit substantial volatility ranging from approximately -23% during the 2008 global financial crisis to +25% in the subsequent 2009 recovery, with additional notable fluctuations including negative returns in 2002, 2011, and 2018, and strong positive performance in years such as 2019 and 2021. This volatility pattern reflects the fund's globally diversified portfolio composition, particularly its exposure to equity risk premiums, which make returns highly sensitive to international market conditions. In sharp contrast, Norway's Governance Quality score remains relatively stable, fluctuating within a narrow range between approximately 1.90 and 2.50 on the WGI scale. Whilst its movements vaguely follow

the general movements of the financial returns, they are largely modest in comparison and occur entirely within the upper tier of global governance quality ratings.

Calculating the Pearson correlation coefficients (r) between the variables for Legitimacy (l), Host Country Governance Quality (q), and Portfolio Stabilisation (DV) based on data from 1999 to 2022, which are the recorded dates for the NBIM's net transfers to/from its budget.

Relationship	Variables Correlated	Pearson Correlation (r)
l v.s. DV (Legitimacy & Stabilisation)	Absolute Deviation from 3% Rule v.s. Annual	-0.1257
q v.s. DV (Governance Quality & Stabilisation)	WGI Composite Index v.s. Annual Return (%)	0.2512
l v.s. q (Legitimacy & Governance Quality)	Absolute Deviation from 3% Rule v.s. WGI Composite Index	-0.0270

Table 1. Pearson correlation coefficients between the operationalised variables

Thirdly, we test our second dependent variable, intergenerational wealth transfer, by plotting the GPFG's total market value growth rate over the same period:

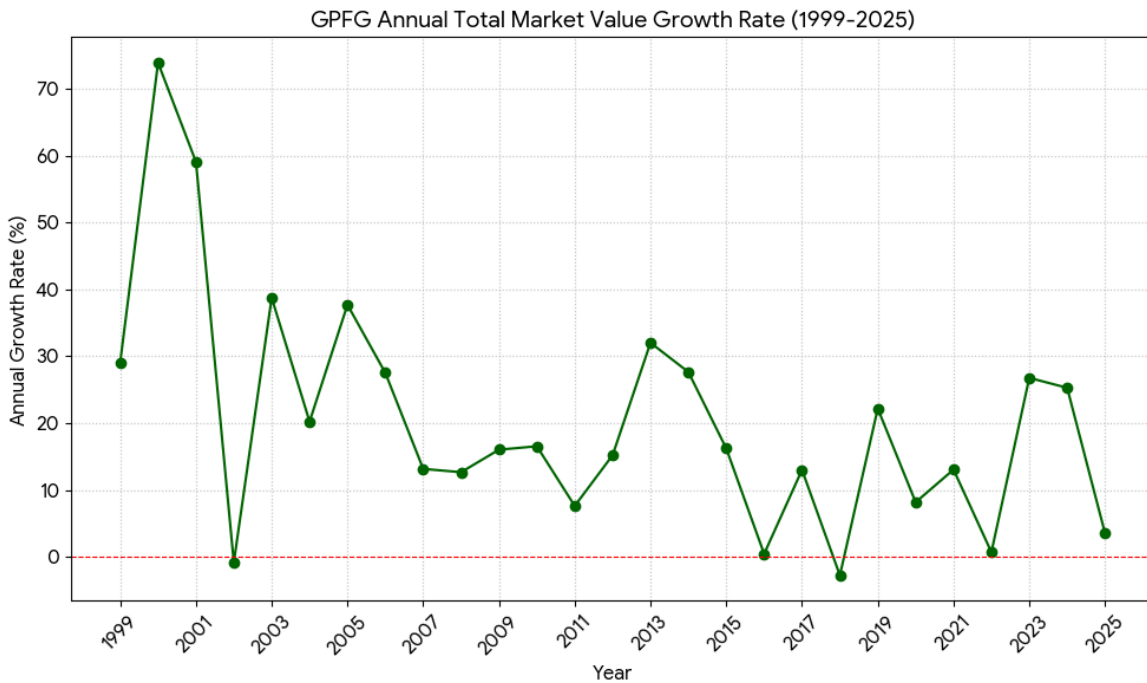


Figure 2. Time Series of the GPFG's Annual Growth Rate of its Market Value (1999-2025). Data from NBIM.no.

From the figure, we can notice some basic observations. The graph is decently volatile, with growth ranging from near-zero, and even negative, in downturns to peaks exceeding 70% at around 1999-2000. Despite downturns, it is obvious that largely, the fund value's growth rates from year to year are positive, thus supporting intergenerational sustainability of the fund's investments.

5 Discussion

Figure 1 supports a core proposition of H1: transparency and governance quality do not mechanically produce superior returns in any given year, but rather operate through indirect pathways affecting long-term strategic consistency, political insulation, and stakeholder legitimacy. The role of transparency-driven legitimacy is therefore not to smooth annual returns directly, but rather to ensure the fund can maintain its mandate, strategic asset allocation, and fiscal rule discipline over a longer horizon amidst unavoidable market cycles and politically motivated deviations. Notably, this dynamic is most prominent during the 2008-2009 period. During 2008, the GPFG experienced its largest single-year loss of approximately -23% as global equity markets collapsed, demonstrating that even the world's most transparent and well-governed SWF cannot insulate portfolio returns from systemic market crises. The governance quality score during this time, although at its lowest, still remained elevated. Subsequent dramatic recovery in 2009, when the fund achieved approximately 25% returns, reflects the benefits of this governance-enabled discipline. Therefore, this illustrates that legitimacy's primary function is protecting long-term strategy during stress rather than preventing drawdowns per se. Similarly, during the 2014-2016 oil price slump, the GPFG maintained its equity-oriented allocation and operated within its fiscal framework, whereas several commodity-backed funds faced ad hoc withdrawals or mandate drift under political pressure, as seen with Algeria's Revenue Regulation Fund (FRR) (Venables, 2016). Opaque reporting during this period, such as with Venezuela's Macroeconomic Stabilization Fund (FIEM), also led certain non-transparent funds to suffer from lack of insulation and resulting macroeconomic instability.

Table 1 demonstrates that there exists a weak negative correlation between fiscal rule adherence and annual returns ($r = -0.1257$), governance quality shows modest positive correlation with returns ($r=0.25$) and fiscal rule adherence and governance quality are essentially uncorrelated. All three correlations are small and likely not statistically significant with only 24 annual observations, confirming that annual returns are dominated by global market conditions. These results strongly support H1's prediction that governance quality has no strong direct relationship with annual returns. The modest positive correlation between governance and returns ($r=0.25$) is reasonable: better governance provides institutional stability, but the small magnitude implies that governance cannot shield the fund from market volatility. This further echoes our result from

Figure 1. The weak negative correlation between fiscal adherence and returns ($r=-0.13$) also implies that following the fiscal rule and discipline does not boost returns in a given year, but instead creates long-term stability. Years with very high returns, such as 2009's spike of +25%, supports this, as when governments respond to crises, it coincides with fiscal rule deviations. The near-zero correlation between fiscal adherence and governance quality ($r=-0.03$) also makes sense because Norway's governance quality remains consistently high while fiscal policy varies with specific economic circumstances. This, however, does not utterly discard these variables as irrelevant, but rather also confirm the long-term argument H2 posits. Effects work through long-term mechanisms like crisis resilience and cumulative wealth building.

From the standpoint of our transparency-legitimacy-economic outcomes framework, Figure 2 supports two of our hypotheses. First, as expected from H1, there is no reason to anticipate a huge correlation between transparency (or governance quality) and annual total market value growth; the latter is driven by multi-factor dynamics. The graph's volatility in this aspect supports the fact that it reflects the combined effects of net petroleum inflows, investment returns, and fiscal withdrawals. This validates our focus on targeted metrics: return consistency and volatility, rather than headline growth. Secondly, consistent with H2, the relative stability of growth rates in more recent years, despite shocks, suggests that legitimacy (proxied by fiscal rule adherence) moderates the volatility of the fund's aggregate position by smoothing fiscal flows and allowing portfolio strategy to remain long-term oriented. When transparency sustains legitimacy, investment policy can be insulated from short-term political pressures, enabling the fund to survive through market cycles, rather than being forced into suboptimal events. This is also consistent with our analysis from Figure 1.

6 Conclusion

This paper examined whether the exceptional transparency of Norway's Government Pension Fund Global (GPFG) contributes to the fund's core economic objectives: long-term financial performance, intergenerational wealth preservation, and macroeconomic stabilisation. Building upon the framework of transparency-legitimacy-outcomes, we argued that transparency primarily operates through political and economic legitimacy to enable policy stability, operational insulation, and long-term viability, rather than directly boosting annual returns.

Time series evidence showed that GPFG returns are dominated by global market dynamics and equity risk (e.g. 2008) followed by recoveries (e.g. 2009). Namely, transparency-driven governance can support modest persistent advantages, such as discipline, resilience, and reduced agency costs, but cannot overturn market efficiency. In other words, we find that transparency is necessary for a fund's durability, but not sufficient for outperformance. Furthermore, we find that legitimacy is the key mediator between transparency and economic effectiveness. Transparency generates domestic and external legitimacy, which protects the fund's strategic asset allocation and adherence to fiscal rules in the face of shocks and political pressures. Finally, we illustrated

that intergenerational wealth is cumulative and contingent to the fund's institutional structure. Although the fund's market value growth rate is volatile, the long-term trend demonstrates sustained wealth accumulation. The GPF operationalises the Hartwick-Solow principle of transforming exhaustible resource rents into a durable financial asset for future generations.

Future work in this area could likely work on a better contextualisation of the case of the Norwegian GPFG against the global backdrop, comparing its transparency scores and relations to governance and economic outcomes to other sovereign wealth funds. Whilst this study adopted a single-case, time-series design to trace mechanisms in the GPFG, it limits external generalisability. Our results may only relate to how findings from the case of the Norwegian SWF may apply to a "best-case" scenario of sovereign wealth funds operating under set metrics of intended transparency and political control. We note here that a more comprehensive dataset constructed by the International Forum of Sovereign Wealth Funds (IFSFW) does exist tracking such metrics of various reputable sovereign wealth funds, but its restrictive nature prevents this study from accessing it. Should such a dataset be used, we can expect that the scope of this research may expand to sharpen the argument of transparency's mediating role and boundary conditions. Comparative designs could also explore the transparency-legitimacy-outcome framework in funds with developmental mandates or in less democratic contexts, where domestic discretion and external legitimacy may conflict.

Collectively, the results from this paper suggest that the GPFG's transparency is effective not because it yields higher returns on demand, but because it aligns societal expectations with long-horizon investing, disciplines political behavior, and reduces governance frictions that otherwise erode value. Simply put: transparency buys patience -- and patience powers long-term economic success.

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